EMBEDDING A SYSTEM OF ABELIAN SHEAVES INTO A SYSTEM OF INJECTIVE SHEAVES OVER A COLIMIT OF SITES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete construction; the compatibility of the resulting maps of systems is verified through functoriality.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Colimit of an inverse system of sites and the functors between them. Sites, the construction of the colimit site and the computation of pullbacks along the transition functors, both cited in the statement. Supplies the sites and functors appearing in the statement.
- Functorial injective embeddings for abelian sheaves on a site. Injectives, the existence of a functorial embedding of an abelian sheaf into an injective one. Supplies the construction used at each index.

Source reference. Original lemma 0A34: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2858–2858. Navigation pointer invoked at proof line 2858; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09YL: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2860–2860. Navigation pointer invoked at proof line 2860; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0A35: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2861–2861. Navigation pointer invoked at proof line 2861; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0730: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2257–2257. Navigation pointer invoked at proof line 2899; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0137: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2900–2900. Navigation pointer invoked at proof line 2900; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered index category and let $(\mathcal{C}_i, f_a)$ be an inverse system of sites over $\mathcal{I}$ as in Sites, Situation 0A34. Set $\mathcal{C} = \operatorname{colim} \mathcal{C}_i$ as in Sites, Lemmas 09YL and 0A35. Moreover, assume given*

(1) *an abelian sheaf $\mathcal{F}_i$ on $\mathcal{C}_i$ for all $i \in \mathrm{Ob}(\mathcal{I})$,*
(2) *for $a : j \to i$ a map $\varphi_a : f_a^{-1}\mathcal{F}_i \to \mathcal{F}_j$ of abelian sheaves on $\mathcal{C}_j$*

such that $\varphi_c = \varphi_b \circ f_b^{-1}\varphi_a$ whenever $c = a \circ b$. Then there exists a map of systems $(\mathcal{F}_i, \varphi_a) \to (\mathcal{G}_i, \psi_a)$ such that $\mathcal{F}_i \to \mathcal{G}_i$ is injective and $\mathcal{G}_i$ is an injective abelian sheaf.

Proof. For each i we pick an injection $\mathcal{F}_i \to \mathcal{A}_i$ where $\mathcal{A}_i$ is an injective abelian sheaf on $\mathcal{C}_i$. Then we can consider the family of maps

$$\gamma_i : \mathcal{F}_i \longrightarrow \prod\nolimits_{b:k\to i} f_{b,*}\mathcal{A}_k = \mathcal{G}_i$$

where the component maps are the maps adjoint to the maps $f_b^{-1}\mathcal{F}_i \to \mathcal{F}_k \to \mathcal{A}_k$. For $a : j \to i$ in $\mathcal{I}$ there is a canonical map

$$\psi_a : f_a^{-1}\mathcal{G}_i \to \mathcal{G}_j$$

whose components are the canonical maps $f_b^{-1}f_{a\circ b,*}\mathcal{A}_k \to f_{b,*}\mathcal{A}_k$ for $b : k \to j$. Thus we find an injection $(\gamma_i) : (\mathcal{F}_i, \varphi_a) \to (\mathcal{G}_i, \psi_a)$ of systems of abelian sheaves. Note that $\mathcal{G}_i$ is an injective sheaf of abelian groups on $\mathcal{C}_i$, see Lemma 0730 and Homology, Lemma 0137. This finishes the construction. □